

Miles Education: Equipping students and professionals for better work opportunities with App Engine

Tell us your challenge. We're here to help.

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ABOUT MILES EDUCATION



About Miles Education

With App Engine
Firestore, Miles E
able to provide a
CPA learning exp
meet its mission
upskilling profes
high-demand dor

Miles Education is on a mission to equip and upskill students and professionals to be future-ready by enabling their career progression. Miles offers a range of industry-recognized certifications including the Certified Public Accountant (CPA) qualification, which allows professionals in India to apply for jobs in some of the largest accounting firms globally.

Industries: Education

they can be more ready.

Location: India

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud results

- Frees up DevOps' time so they can experiment with new ideas
- Reduces new application deployment from 3 hours to under 2 minutes
- Supports 12x growth of the company with high scalability
- Saves cost and reduces waste with

Deploy
application
100 seconds

searce

About Searce

One of the largest Google Cloud partners, Searce is "futurifying" businesses by pushing the boundaries of tech innovation with end-to-end design industry solutions for its clients all around the world.

Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace
(<https://workspace.google.com/>)

Calendar
(<https://workspace.google.com/products/calendar/>)

the
flexibility of
Google
Cloud

Docs

(<https://workspace.google.com/products/docs/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

In the finance sector, being a Certified Public Accountant (CPA) can spearhead one's career growth. However, before 2010, it was almost impossible to get a CPA qualification in India as there weren't any organizations offering the course.

To address that need, Varun Jain, a CPA, Certified Management Accountant (CMA), and Harvard Business School alumnus, co-founded Miles Education (<https://mileseducation.com/cpa>) with his wife Meenakshi Jai, a Stanford Business School alumnus to enable more students and professionals across the country to be CPA qualified. While Varun focused on building the product, Meenakshi was the brains behind marketing and sales. "When we first started, our annual target was around 1,100 to 1,500 students annually," explains Shivam Chopra, Chief Technology Officer (CTO) at Miles Education. However, when the pandemic struck in 2020, the organization saw a sudden influx of students who were looking to upskill themselves during the lockdowns. New student enrollment went up by 12 times, from 1,500 annually to 1,500 every month.

"We launched with a very low-level learning management system (LMS), and leveraged the managed services that came with the website hosting provider," says Chopra. "However, the problem was that with a LMS, we required a lot of

APIs and that was tough to handle because of the high latencies it caused, and we simply couldn't scale."

Beyond just the LMS, Miles Education was also creating new training platforms, which required additional hosting. As the team was still working out of a shared office space, Chopra decided to onboard a cloud provider that was more well-known at the time, which worked well. This was until the website hosting costs escalated.

"As a bootstrapped startup, cost really mattered to us and the alarm bells started ringing when we saw how much we were paying every month just to keep our services up and running. We started to look around to try different cloud providers, and found Google Cloud (<https://cloud.google.com/>) to be the most cost-efficient," says Chopra. "We used Firebase a lot, and realized that Firebase was part of Google Cloud, which further sealed our decision."

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Scaling the business quickly with App Engine automation

Cost may have been the initial push for Miles Education to embark on its Google Cloud journey, however, Chopra soon discovered the power of App Engine (<https://cloud.google.com/appengine/>) as the key to a much quicker deployment.

"I call it the 100 seconds deployment," he says, explaining that with just a few sets of commands,

App Engine was able to build, deploy, and load balance, without requiring any intervention. What used to take the team three hours to deploy can be done in 100 seconds, so the DevOps team at Miles Education were able to spend its time building and experimenting on more innovative features.

Another advantage of App Engine is its command line capabilities, instead of a Dockerized structure, as this meant that engineers could easily adjust and tweak the codes more conveniently when required.

"Very honestly though, there was a huge learning curve with Google Cloud, such as when I didn't realize we needed to enable load balancers separately, since the previous cloud provider had it all done by default," explains Chopra.

That said, Chopra and the team now understand the value that this flexibility brings. "In hindsight, now that I understand the product better, I realized that the way Google Cloud is set up works for us as we aren't getting billed for the things we don't use."

Because of the way Google Cloud is set up, Chopra feels more confident with security as he is able to control exactly which ports and services he needs to access or give permission for. This is unlike other tools that are Dockerized, and therefore less transparent.

"Even if I want to internally communicate between one server to another without giving a direct host access to the primary server, I can do that with Google Cloud," says Chopra. The team also uses

Serverless VPC

(<https://cloud.google.com/vpc/docs/serverless-vpc-access>)

for added data security.



Boosting developer confidence with a trusted partner

Miles Education now works with Searce

(<https://www.searce.com/>), which provides support to the team where required. The high level of commitment Searce presents was evident when Miles Education experienced mining in its servers.

"We once discovered mining on our servers, and Searce helped us connect with the Google team to get its DevOps team working to solve the issue. They also did not hesitate to waive the fee entirely for us that month," says Chopra.

With a team that's constantly modifying and updating applications, and experimenting with new ideas, Chopra is well aware that no platform is completely secure, particularly since it uses a lot of open source tools. However, knowing that it has

Searce as a reliable partner has given the team enough confidence to build and explore freely.

"Searce, as our technology partner, plays a crucial role alongside Miles Education, aiding our organization in enhancing processes through the use of Google services and providing support to our team," says Jain.

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Building on a wider Google Cloud ecosystem

Beyond App Engine and Firebase, Miles Education uses Compute Engine

(<https://cloud.google.com/compute>) to run its applications and Cloud SQL

(<https://cloud.google.com/sql>) for its database. Its DevOps team also leverages Cloud Build for the build and deployment of containerization products and is exploring more use cases of Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>).

Being Google Workspace users, the most-used product for Miles Education is Sheets

(<https://workspace.google.com/products/sheets/>), as Chopra says that it is the starting point of everything it does at the organization. This is followed by Docs (<https://workspace.google.com/products/docs/>). "Having multiple team-members working on a document at the same time, with everything being updated has been amazing for our workflow, as it avoids duplication of files and versioning, therefore eliminating any confusion when we collaborate."

With Calendar

(<https://workspace.google.com/products/calendar/>) having been built into the platform, visitors to the website can already set up an appointment easily to speak with a customer service representative.

However, Miles Education is experimenting with artificial intelligence (AI) to build a machine learning platform that can further support sales. "I want to build a tool where students can ask any question, and get an answer immediately. They should also be able to submit their existing qualifications and find out if they are equipped to undertake the CPA or any other certification they wish to take. I look forward to learn more about the tools that Google has in store for us," explains Chopra.

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